| Defense (gem5 build) | Criterion source | Counts as a transmission | Does not count | Trace signal | Window |
|---|---|---|---|---|---|
| STT-Spectre `/work/stt` scheme 2, Spectre threat model | `stt.json`: `transmitters:[ld,br_x]`, `checkers.ld: check_ld.py` *No criterion JSON: generic `check_ld` gate.* | The transmitter load executes in $W$ and obtains data: it sends a read packet (`reached_cache`) or, sending none, completes (`store_forward`). | Load never executes; or executes, sends no packet and never completes. | `LSQUnit: Executing load, successfully sent out packet(s)`; `O3PipeView: complete`. | `Executing load` tick in $W$. |
| Recon-STT `gem5-recon-modded` scheme 2 | `recon.json`: same keys as `stt.json`. *No criterion JSON: generic `check_ld` gate.* | Same gate, but this build never prints the packet line, so only `store_forward` (the load completes) can fire. | Delay-unit bounce; also a load that sent a packet but was squashed before completing (gate bug). | `LSQUnit: Executing load`; the real send, `Memory request... was sent`, is not parsed; `O3PipeView: complete`. | `Executing load` tick in $W$. |
| InvisiSpec `amulet-InvisiSpec --ruby`, `SpectreSafe InvisibleSpec` | `invisispec.json`: `leak.form:exists`, `structures: [L1D,L2,DIR,DTLB]`, `kinds:[fill,evict, inv,coh,repl_touch]` | Any such event caused by any squashed instruction younger than lc (not only the transmitter). | SB, LLCSB `buf_fill/buf_release`; `expose/validate`; MSHR/TBE; L1I, ITLB. | SimSpect hook lines `O3PipeView:simspect: <struct>:<kind>` (Ruby `CacheMemory`, TLB, SLICC transitions). | Event tick in $W$; Futuristic mode: any instruction, before its visibility point. |
| SpecLFB `amulet-SpecLFB` classic caches, `--l2cache`, `--scheme=Speclfb` | `speclfb.json`: `leak.form:exists`, `structures:[L1D,L2]`, `kinds:[fill,evict]` | An L1D or L2 `fill`, or the `evict` of a victim, caused by a squashed instruction younger than lc. | Parking in the LFB (`buf_fill/buf_release`); hits, `repl_touch`; DTLB, L1I; MSHRs. | SimSpect hook lines (classic `BaseCache` allocate and evict, `refillBlocks`). | Event tick in $W$. |
| CleanupSpec `amulet-CleanupSpec --ruby`, `Cleanup_FOR_L1L2` | `cleanupspec.json`: `leak.form: per_line_balance`, `undo_obligations: L1D:[install,victim], L2:[install]` | Per structure and line, squashed installs minus `cleanup_inv`, or L1D victims minus `cleanup_restore`, is nonzero (too little or too much undo; skips and drops are not undo). | Fully undone footprint; a fill evicted again inside $W$; L2 victims, replacement state, DIR; cleanup timing (unXpec). | SimSpect hook lines: `CacheMemory` plus cleanup events (`Inv_CLEANUP`, `Load_ROLLBACK`, LSQ `cleanup_done`). | Footprint in $W$; undo from the squash to `cleanup_done`. |

$W$: `lc_retire` $< t <$ `fnc_retire`, and every branch the model marks unresolved is still unresolved at $t$ (SimSpect `BR:resolve_visible` hook line). A specified lc that never retires disqualifies the test; a run that did not realize the modeled speculation (mispredict, stall, forced prediction) is invalid, never a hit.
□ defense-specific criterion JSON